Mathematical Model and Lumped RLC Formulation for the Simscape Implementation of a Shielded Three-Phase Cable

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Abstract

This document develops the mathematical model required to implement a shielded three-phase power cable in Simscape (MATLAB) as a lumped RLC network. The multiconductor transmission-line (MTL) formulation is reduced to per-unit-length (p.u.l.) series and shunt matrices, then mapped onto single- and multi-section π models. A complete numerical dimensioning is carried out for the reference case of a copper cable with 90 mm^2 per phase, collectively shielded, of length 120 m. The section-count criterion for the cascaded π model is derived from the propagation velocity and the target bandwidth, and the shield/common-mode representation is included so that the same model serves both fundamental-frequency studies and high-frequency phenomena (reflected-wave overvoltage, common-mode current). References follow IEEE practice.

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1 Scope and modeling objectives

The objective is a circuit-equivalent cable model suitable for time-domain simulation in Simscape / Simscape Electrical (Specialized Power Systems). Three requirements drive the formulation:

1. **Fidelity at the fundamental and low harmonics** for steady-state and power-quality studies (voltage drop, charging current, losses).
2. **Representation of the metallic shield** as an explicit conductor, so that phase-to-shield capacitance and shield bonding can be analyzed, which is the prerequisite for common-mode and EMC studies.
3. **Controlled bandwidth** through a cascaded π structure whose number of sections is chosen for the highest frequency of interest, allowing the same topology to capture cable resonances and reflected-wave behavior at PWM-converter output.

The cable is treated as a uniform, laterally homogeneous line. End effects, joints and terminations are excluded; they are added as separate lumped elements when needed.

2 Cable construction and reference data

The reference cable is a three-core copper construction with a collective metallic shield, representative of a 0.6/1 kV EMC drive cable. The assumed geometry is summarized in Table 1. These values define the worked example; for a specific product the manufacturer datasheet p.u.l. values should be substituted directly into Sections 5–7.

Table 1: Reference geometry and material data.

Quantity	Symbol	Value
Phase conductor cross-section	A_c	90 mm^2 (Cu, stranded)
Effective conductor radius	r_c	6.0 mm
Geometric mean radius	GMR	$0.77\,r_c \approx 4.6$ mm
Insulation (XLPE) rel. permittivity	ε_r	2.5
Insulation loss tangent	$\tan\delta$	4×10^{-3}
Radius over core insulation	r_i	8.0 mm
Core lay-up	—	trefoil, touching
Center-to-center spacing	S	16 mm
Shield inner radius	R_s	18 mm
Shield cross-section (Cu)	A_s	16 mm^2
Cable length	ℓ	120 m
Copper resistivity (20 °C)	ρ_{20}	$1.724 \times 10^{-8}\ \Omega\,\mathrm{m}$
Temperature coefficient	α	$3.93 \times 10^{-3}\ \mathrm{K}^{-1}$

3 Distributed-parameter formulation

With phase conductors a, b, c and shield s, the cable is a four-conductor MTL. In the frequency domain the telegrapher equations are

$$-\frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}x} = \boldsymbol{Z}\,\boldsymbol{I}, \qquad -\frac{\mathrm{d}\boldsymbol{I}}{\mathrm{d}x} = \boldsymbol{Y}\,\boldsymbol{V}, \tag{1}$$

with the p.u.l. series impedance and shunt admittance matrices

$$\boldsymbol{Z} = \boldsymbol{R} + j\omega\boldsymbol{L}, \qquad \boldsymbol{Y} = \boldsymbol{G} + j\omega\boldsymbol{C}. \tag{2}$$

$\boldsymbol{V}$ collects conductor-to-reference voltages and $\boldsymbol{I}$ the conductor currents. For a symmetric three-phase cable the 3×3 phase blocks have the balanced structure

$$\boldsymbol{Z}_{abc} = \begin{bmatrix} Z_s & Z_m & Z_m \\ Z_m & Z_s & Z_m \\ Z_m & Z_m & Z_s \end{bmatrix}, \qquad \boldsymbol{Y}_{abc} = \begin{bmatrix} Y_s & Y_m & Y_m \\ Y_m & Y_s & Y_m \\ Y_m & Y_m & Y_s \end{bmatrix}. \tag{3}$$

Shield elimination (Kron reduction). When the shield is bonded to ground at both ends, its voltage is constrained and the four-conductor system reduces to an effective 3×3 phase model:

$$\boldsymbol{Z}_{abc}^{\mathrm{eff}} = \boldsymbol{Z}_{pp} - \boldsymbol{Z}_{ps}\, Z_{ss}^{-1}\, \boldsymbol{Z}_{sp}, \tag{4}$$

where pp, ps, sp, ss are the phase, phase-shield, shield-phase and shield partitions of the full 4×4 matrix. Single-point bonding suppresses the shield circulating current and is treated by keeping the shield as an explicit floating conductor instead of applying (4). The lumped model below retains the shield explicitly so that either bonding scheme can be imposed in Simscape.

Modal / sequence decomposition. The balanced structure (3) is diagonalized by the symmetrical-component (or, in real arithmetic, Clarke) transform, giving decoupled positive/negative and zero modes

$$Z_1 = Z_s - Z_m, \qquad Z_0 = Z_s + 2Z_m, \qquad Y_1 = Y_s - Y_m, \qquad Y_0 = Y_s + 2Y_m. \tag{5}$$

Equation (5) is the bridge between the physical p.u.l. quantities and the positive-/zero-sequence parameters used by the Simscape three-phase line block.

4 Per-unit-length parameters

4.1 Series resistance

The DC resistance of one conductor is

$$R'_{\mathrm{dc}}(\theta) = \frac{\rho_{20}}{A_c}\big[1 + \alpha(\theta - 20)\big]. \tag{6}$$

The AC resistance adds skin (y_s) and proximity (y_p) corrections following IEC 60287 / IEEE cable-loss practice:

$$R'_{\mathrm{ac}} = R'_{\mathrm{dc}}\,(1 + y_s + y_p), \qquad y_s = \frac{x_s^4}{192 + 0.8\,x_s^4}, \qquad x_s^2 = \frac{8\pi f}{R'_{\mathrm{dc}}}\,10^{-7}\,k_s, \tag{7}$$

with $k_s \approx 1$ for round stranded conductors. At 50 Hz the correction is a few percent for 90 mm^2; at the kHz range relevant to converter harmonics it grows and is captured by re-evaluating (7) at the dominant harmonic, or by a Foster (R–L ladder) branch when broadband fidelity is required.

4.2 Series inductance

The positive-sequence (loop) inductance per phase is

$$L_1' = \frac{\mu_0}{2\pi}\ln\frac{\mathrm{GMD}}{\mathrm{GMR}}, \qquad \mathrm{GMD} = \sqrt[3]{S_{ab}S_{bc}S_{ca}} = S \ \text{(trefoil)}. \tag{8}$$

The zero-sequence inductance depends on the common-mode return path. With the shield as the dominant return at radius R_s,

$$L_0' \approx \frac{\mu_0}{2\pi}\ln\frac{R_s}{\mathrm{GMR}} \ + \ \Delta L_{\mathrm{earth}}, \tag{9}$$

where $\Delta L_{\mathrm{earth}}$ accounts for the earth-return contribution (Carson/Pollaczek) when the shield is grounded at both ends. L_0' is therefore bonding-dependent and must be re-evaluated for the actual installation.

4.3 Shunt capacitance

Two physical capacitances describe the shielded cable: the core-to-shield capacitance C_g' and the core-to-core capacitance C_m'. For an individually screened core the former is the coaxial value

$$C_g' = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(r_i/r_c)}, \tag{10}$$

while for a collective shield C_g' and C_m' are obtained from the Maxwell potential-coefficient matrix $\boldsymbol{P} = \boldsymbol{C}^{-1}/(2\pi\varepsilon)$ of the core/shield geometry. The sequence capacitances follow from (5):

$$C_1' = C_g' + 3\,C_m', \qquad C_0' = C_g'. \tag{11}$$

C_1' is the operating (line) capacitance that sets the charging current; C_0' is the capacitance seen by the common mode.

4.4 Shunt conductance

Dielectric losses give

$$G' = \omega\,C'\tan\delta, \tag{12}$$

negligible at 50 Hz but retained as a small parallel conductance to damp the high-frequency π-network resonances and avoid undamped numerical oscillation.

4.5 Shield parameters

The shield DC resistance is $R_s' = \rho_{20}/A_s$; at high frequency the braid contributes a transfer impedance, but for the lumped model the shield is represented by its series R_s', its self-inductance, and the C_g' coupling to each core.

5 Numerical dimensioning of the reference cable

Applying (6)–(12) to Table 1 gives the p.u.l. values in Table 2. The conductor AC resistance is reported at the 90 °C rated temperature with a 50 Hz skin/proximity correction of about 2%.

Lumped totals for $\ell = 120$ m. Multiplying by ℓ gives the total elements of the equivalent network:

$$\begin{aligned}
&R = R_{\mathrm{ac}}'\,\ell = 0.248 \times 0.120 = 29.8\ \mathrm{m\Omega}\ \text{per phase},\\
&L_1 = L_1'\,\ell = 30.0\ \mu\mathrm{H}\ \text{per phase}, \qquad L_0 = 32.4\ \mu\mathrm{H},\\
&C_1 = C_1'\,\ell = 72\ \mathrm{nF}\ \text{(operating, line)}, \qquad C_g = C_g'\,\ell = 36\ \mathrm{nF}\ \text{per phase to shield},\\
&R_s = R_s'\,\ell = 130\ \mathrm{m\Omega}\ \text{(shield)}.
\end{aligned}$$

Table 2: Computed per-unit-length parameters of the reference cable.

Parameter	Symbol	Value	Unit
Conductor resistance, DC 20 °C	R'_{dc}	0.192	Ω/km
Conductor resistance, AC 90 °C, 50 Hz	R'_{ac}	0.248	Ω/km
Shield resistance, DC 20 °C	R'_s	1.08	Ω/km
Positive-sequence inductance	L'_1	0.25	mH/km
Zero-sequence inductance (both-end bonding)	L'_0	0.27	mH/km
Core-to-shield capacitance	C'_g	0.30	μF/km
Core-to-core capacitance	C'_m	0.10	μF/km
Positive-sequence capacitance	$C'_1 = C'_g + 3C'_m$	0.60	μF/km
Zero-sequence capacitance	$C'_0 = C'_g$	0.30	μF/km
Shunt conductance (50 Hz)	G'_1	0.75	μS/km

Derived line quantities. The propagation velocity, characteristic impedance and one-way delay of the positive mode are

$$v = \frac{1}{\sqrt{L'_1 C'_1}} = 8.2 \times 10^7 \text{ m/s}, \qquad Z_0 = \sqrt{\frac{L'_1}{C'_1}} \approx 20\ \Omega, \qquad \tau = \frac{\ell}{v} = 1.47\ \mu\text{s}. \tag{13}$$

The first open-end (quarter-wave) resonance of the 120 m cable is

$$f_{\lambda/4} = \frac{v}{4\ell} \approx 170 \text{ kHz}, \tag{14}$$

which sets the bandwidth the lumped model must reach to reproduce the first resonance and the reflected-wave behavior at a converter output.

6 Lumped RLC model

6.1 Single π section

For each phase the elementary nominal-π cell places the total series branch $R + j\omega L$ between two halves of the total shunt branch, $Y/2$ at each end. Including the shield explicitly, the shunt branch is split into the core-to-shield capacitance C_g and the core-to-core capacitance C_m, and the shield node carries R_s, L_s to its bonding point. The single-π cell is accurate while the cable is electrically short, i.e. $\ell \ll \lambda/10$. From (13) this holds up to

$$f_{\max}^{(1\pi)} = \frac{v}{10\,\ell} \approx 68 \text{ kHz}, \tag{15}$$

adequate for fundamental and low-harmonic studies but not for the resonance in (14).

6.2 Cascaded N-section π model

For broadband fidelity the cable is split into N identical π cells, each representing a length ℓ/N with element values R/N, L/N, C/N, G/N. The number of cells is set by the highest frequency of interest $f_{\max}$ using a "cells-per-wavelength" rule. Requiring k cells per shortest wavelength $\lambda_{\min} = v/f_{\max}$,

$$\boxed{N \geq k\,\frac{\ell\, f_{\max}}{v}}, \qquad k \approx 8. \tag{16}$$

For the reference cable, Table 3 lists the resulting section count.

Table 3: Required number of π sections for the 120 m reference cable ($k = 8$).

Target bandwidth $f_{\max}$	$\lambda_{\min} = v/f_{\max}$	N
10 kHz	8.2 km	1
100 kHz	820 m	2
500 kHz	164 m	6
1 MHz	82 m	12
3 MHz	27 m	35

A practical choice for PWM-drive studies (reflected wave, dv/dt stress) is $N = 20$–40, since the relevant edge spectra extend into the low-MHz range. The trade-off is computational: each added section introduces one more L–C state pair and lowers the simulation step required for stability.

6.3 Phase-domain versus sequence-domain realization

Two equivalent realizations of each cell are possible:

- **Sequence-domain:** use R_1, L_1, C_1 and R_0, L_0, C_0 from Table 2. Valid for the balanced, transposed assumption and directly accepted by the Simscape three-phase π-line block.
- **Phase-domain:** populate the 3×3 (or 4×4 with shield) $\boldsymbol{R}, \boldsymbol{L}, \boldsymbol{C}$ matrices from (3). This retains explicit inter-phase and phase-shield coupling and is preferred for unbalanced or common-mode studies.

6.4 Shield and common-mode representation

The common-mode loop is the three phases in parallel returning through the shield (and earth). Its lumped parameters are obtained from the zero-sequence quantities: the equivalent CM inductance is $L_0/3$ in series with the shield R_s, L_s, and the CM capacitance is $3C_g$. Retaining the shield node with selectable bonding (both-end, single-end, or floating) lets the same model reproduce shield circulating current and the high-frequency CM path that governs bearing current and conducted EMI in drive applications.

7 Simscape implementation

The model maps directly onto Simscape Electrical components.

1. **Ready-made block.** The *Three-Phase PI Section Line* (Specialized Power Systems) accepts the positive- and zero-sequence R_1, R_0 (Ω/km), L_1, L_0 (H/km), C_1, C_0 (F/km), the length, and the number of sections N. Enter the values of Table 2, $\ell = 0.12$ km, and N from Table 3. This is the fastest route when an explicit shield node is not needed.
2. **Explicit shield / custom cell.** When the shield must be accessible, build one π cell from foundation components — *Resistor*, *Inductor* (or mutual-inductor / *Three-Phase Mutual Inductance Z1-Z0* for coupling), and *Capacitor* — with the per-cell values R/N, L/N, C_g/N, C_m/N, G/N, and a dedicated shield node carrying R_s/N, L_s/N. Cascade N cells and expose the shield terminal for bonding.
3. **Numerical settings.** The cascaded π network is stiff; use a variable-step stiff solver (`ode23t`/`ode15s`) or a fixed-step discrete solver with $T_s \lesssim 1/(20 f_{\max})$. The small dielectric conductance G from (12) damps the inter-cell resonances and improves conditioning.

4. **Parameterization.** Drive the block masks from a MATLAB script that computes Table 2 from the geometry of Table 1, so the same model regenerates for any length or cross-section.

8 Validation

The lumped model is validated against the closed-form distributed line over the intended bandwidth:

1. **Input impedance.** Compare the open- and short-circuit driving-point impedance of the N-cell network with the exact $Z_{\text{in}} = Z_0 \coth(\gamma\ell)$ and $Z_0 \tanh(\gamma\ell)$, where $\gamma = \sqrt{\boldsymbol{Z}\boldsymbol{Y}}$. The first resonance must match (14) within a few percent.

2. **Charging current.** At 50 Hz, the no-load current $I_c = \omega C_1 \ell\, U_{\text{ph}}$ must equal the analytic value.

3. **Step / reflected wave.** For a fast voltage edge at the sending end, the receiving-end overshoot and its rise must approach the lossless prediction set by τ and the terminal reflection coefficient $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$ as N increases.

Convergence of these three indicators with increasing N confirms that the chosen section count is sufficient.

9 Application notes

The same parameters quantify two phenomena common in converter-fed installations:

Reflected-wave overvoltage. With $Z_0 \approx 20\ \Omega$ and a motor input impedance of hundreds of ohms, $\Gamma \to 1$ and the receiving-end voltage can approach twice the DC-link-derived pulse. The critical cable length above which full doubling occurs is $\ell_c = v\, t_r/2$, with t_r the converter edge rise time; for $t_r = 100$ ns, $\ell_c \approx 4$ m, so the 120 m reference cable is firmly in the long-cable regime.

Common-mode current. The $3C_g = 108$ nF shield path, excited by the converter common-mode voltage, sets the magnitude and spectrum of the common-mode current returning through the shield bonds. Single-point versus both-point bonding is studied directly by changing the shield termination in the model.

References

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A Parameter summary for Simscape

Table 4: Direct entries for the Simscape three-phase π-line block ($\ell = 0.12$ km).

Field	Positive sequence	Zero sequence
R (Ω/km)	0.248	$0.248 + 3R'_s = 3.49$ [1]
L (mH/km)	0.25	0.27
C (μF/km)	0.60	0.30
Length (km)	0.12	
Sections N	from Table 3	

[1] The zero-sequence resistance includes the shield/earth return contribution; the exact value depends on the bonding scheme and is finalized after the Kron reduction (4) for the chosen installation.